

Lakshya JEE 2027

Chemistry by Faisal Razaq Sir

Electrochemistry

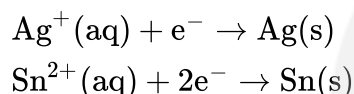
DPP: 4

Total Duration - 25 Min

Q1 E° for the half cell $\text{Zn}^{2+} | \text{Zn}$ is -0.76 e.m.f. of the cell $\text{Zn} | \text{Zn}^{2+} (1\text{M}) || 2\text{H}^+ (1\text{M}) | \text{H}_2 (1\text{atm})$ is:

- (A) -0.76 V
 (B) $+0.76\text{ V}$
 (C) -0.38 V
 (D) $+0.38\text{ V}$

Q2 The standard electrode potentials for the reactions:



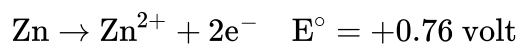
at 25°C are 0.80 volt and -0.14 volt, respectively.

The standard emf of the cell $\text{Sn} | \text{Sn}^{2+} (1\text{M})$

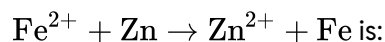
$|| \text{Ag}^+ (1\text{M}) | \text{Ag}$ is:

- (A) 0.66 volt (B) 0.80 volt
 (C) 1.08 volt (D) 0.94 volt

Q3 The standard oxidation potentials E° , for the half reaction are as:



The standard emf of the cell:



- (A) 0.35 volt (B) -0.35 volt
 (C) $+1.17\text{ volt}$ (D) -1.17 volt

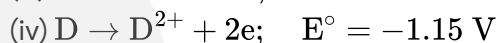
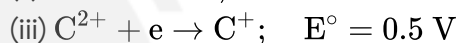
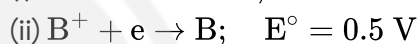
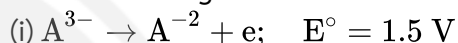
Q4 $E^\circ (\text{Ni}^{2+}/\text{Ni}) = -0.25\text{ volt}$
 $E^\circ (\text{Au}^{3+}/\text{Au}) = 1.50\text{ volt}$

The emf of the voltaic cell.



- (A) 1.25 volt
 (B) -1.75 volt
 (C) 1.75 volt
 (D) 4.0 volt

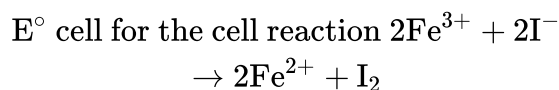
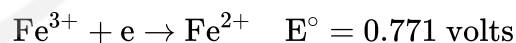
Q5 For the following E° values of half cells



What combination of two half cells would result in a cell with the largest potential?

- (A) (i) and (iii) (B) (i) and (iv)
 (C) (ii) and (iv) (D) (iii) and (iv)

Q6 Given electrode potentials:



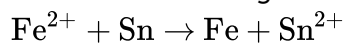
is:

- (A) $(2 \times 0.771 - 0.836) = 1.006\text{ volts}$
 (B) $(0.771 - 0.5 \times 0.536) = 0.503\text{ volts}$
 (C) $0.771 - 0.536 = 0.235\text{ volts}$
 (D) $0.536 - 0.771 = -0.235\text{ volts}$

Q7 The standard reduction electrode potentials of Fe^{2+}/Fe and Sn^{2+}/Sn are -0.44 V and -0.14 V respectively. What will be the standard

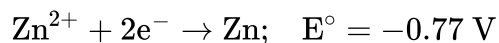
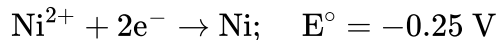


E.M.F. of the following cell reaction



- (A) 0.3 V
(B) -0.58 V
(C) +0.58 V
(D) -0.30 V

Q8 The standard electrode potentials of the two-half cell are given below



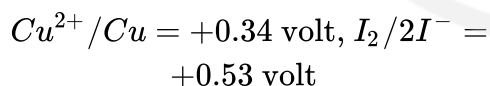
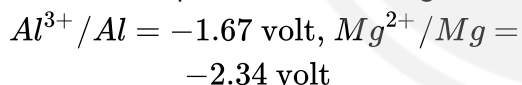
The emf of cell formed by combining the two half cells would be

- (A) -1.02 volt (B) +0.52 volt
(C) +1.02 volt (D) -0.52 volt

Q9 The standard electrode potential value of the elements A, B and C are 0.68, -2.50 and 0.50 V respectively. The order of their reducing power is:

- (A) $A > B > C$
(B) $A > C > B$
(C) $C > B > A$
(D) $B > C > A$

Q10 The reduction potential values are given below



Which one is the best reducing agent?

- (A) Al
(B) Mg
(C) Cu
(D) I_2



Answer Key

Q1 (B)

Q2 (D)

Q3 (A)

Q4 (C)

Q5 (B)

Q6 (C)

Q7 (D)

Q8 (B)

Q9 (D)

Q10 (B)



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